



PHOTO BY **HARDIK CHHABRA**

THE TATA PUSH

India's chip ecosystem gets a major boost as N. Chandrasekaran reveals the group's plans

► **It's not just Vedanta, but also the Tata group that is keen on stepping into the semiconductor business.** Tata Sons Chairman N. Chandrasekaran recently said that the company plans to use a part of its \$90-billion investment over the next five years to set

up ATMP and chip fabs. He also intends to foray into advanced chip manufacturing in a few years. The diversified conglomerate has recently appointed semiconductor industry veteran Randhir Thakur as the CEO & MD of Tata Electronics, a company with exper-

tise in making precision electronics components. The greenfield chip business will also be housed under it, which has a manufacturing facility in the Krishnagiri district of Tamil Nadu. "We will start with manufacturing. Then we'll get into assembly and packaging. The next stage will be semiconductor testing, then fabless, [where] we will do the design. And then, we get into fabs. Those are the stages of integration we will go through, but probably not for a very long period," Chandrasekaran said at a BT event recently.

As semiconductor manufacturing involves a big investment, he has already chalked out a plan. "We have got to make sure that we are doing the right

thing and investing in the right node. And before we have our own architecture, we will have our own electronic/electrical architecture. And this will be for global markets."

He is also looking at acquisitions and collaborations. For instance, in June 2022, Japan-based Renesas Electronics that supplies advanced semiconductors, had tied up with two Tata group firms to develop chips for tech across automotive, IoT and 5G systems. It will also work with TCS to set up a development centre in Bengaluru.

Whatever the node, with Tata bringing its heft into the chips industry, the ecosystem and job opportunities in the Indian market are up for a major boost.

| 89

"To start with, we want to bring those talented people in who have experience in wafer fabs, OSAT, and panel, to form the team. The second part will be to mirror the folks with the hunger, drive, curiosity and problem-solving ability from India, to those with the expertise," says David Reed, CEO of Vedanta-Foxconn Semiconductors. (See *interview for more.*)

BEYOND ENGINEERING

While most roles in the chip industry need an engineering background, there are jobs around maintenance, machinery, supply chain, IT and accounting that need professionals and diploma holders from other streams. Alug of NLB Services, who works with several semiconductor firms globally, says job roles such as operators, technicians, QC (quality check) and supply chain managers, etc., are some of the non-engineering profiles available at a foundry firm.

Also globally, the setting up of a chip manufactur-

ing ecosystem creates an employment opportunity in the ratio of 1:6 for direct and indirect roles. This means, for each fab-related job, six more roles are created in related fields of IT/ ITeS, hardware, software and supply chain, among others.

Consequently, a large number of job opportunities related to design, manufacturing, training and skilling, supply chain management, chemicals and materials engineering, packaging, testing, and logistics are expected to crop up in the next two to five years. Going back to Deloitte's finding of over a million additional skilled workers required by 2030, and the talent crunch only expected to widen further, India—which produces a large number of engineers every year—can substantially leverage its talent advantage. And just like it became the IT hub for the world, the country can aim to become the primary supplier of semiconductor talent, and potentially chips too in some time, to the world. **BT**

@nidhisingal